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QN 1. WHY CREATE AI SYSTEMS THAT PLAY GAMES?

Researchers often use games to develop and test AI because they provide structured, rule-based environments with clear objectives—ideal for evaluating intelligent behaviour (Hu et al., 2023). Many games mirror real-life decision-making but in a controlled, repeatable way, allowing safe exploration of planning, learning, and reasoning.

Historically, game-playing has served as a benchmark for AI. Early breakthroughs like the checkers-playing program by Arthur Samuel in the 1950s demonstrated the potential of machine learning (Wiederhold & McCarthy, 1992). Following that, major milestones—such as IBM's Deep Blue defeating world chess champion Garry Kasparov in 1997—have marked the progression of AI research (Goodrich, 2021).

Modern research continues to build on this tradition. DeepMind's AlphaStar reached grandmaster level in *StarCraft II*, a complex real-time strategy game, using deep reinforcement learning (Vinyals et al., 2019). OpenAI Five showed similar success in *Dota 2*, a highly dynamic multiplayer game requiring coordination and long-term planning (OpenAI et al., 2021).

My own view is that game environments offer a kind of "safe lab" for trial and error. For example, simulated driving games like TORCS offer an ethical advantage in AI research by enabling the training and testing of autonomous agents in realistic, dynamic environments without risking human lives or property. As Loiacono et al. (2010) note, TORCS provides a safe yet lifelike space to explore driving behaviours that would be ethically problematic in real traffic.

Essentially, games are used by researchers in their field for AI systems not only because they are entertaining but also because they are effective abstractions of real-world tasks that enable both theoretical and practical advances in AI.

QN 2. THREE POSSIBLE APPLICATION AREAS FOR AI SYSTEMS THAT PLAY GAMES

2.1 EDUCATION AND TRAINING

Game-playing AI has significant potential in education and training. Zhan et al. (2022) and Leitner et al. (2023) show that game-based learning enhances motivation and helps students grasp complex AI concepts through narrative, challenge, and interactivity. In these settings, AI often acts as an adaptive opponent, adjusting to learners' decisions in real time. Squire (2011) notes that video games foster a participatory learning culture and deserve academic attention for their educational value. By merging AI's adaptability with the immersive nature of games, learning becomes more personalized, interactive, and effective.

2.2 HEALTHCARE

Game-playing AI has strong potential in surgical decision-making, particularly in high-pressure situations like trauma or emergency neurosurgery. In such cases, even experienced surgeons can be limited by stress or incomplete information. Reinforcement learning-based multi-agent systems can assist by analysing data in real time, simulating outcomes, and recommending optimal actions—often faster and more objectively than humans. Acting as adaptive, ever-alert assistants, these systems continuously learn and evolve. As shown by Hashimoto et al. (2018), AI can even predict complications during live procedures. Trained in high-stakes environments, these AI agents mirror the strategic, trial-and-error thinking of expert surgeons under pressure.

2.3 FINANCE

Multi-agent game-playing AI shows strong potential in high-stakes fields like finance, where fast, accurate decisions are essential. Reinforcement learning systems can process vast market data and adapt in real time, often outperforming human traders.

For instance, Zhang et al. (2020) demonstrated how deep reinforcement learning agents can outperform traditional models in portfolio management by continuously adapting to dynamic environments. Additionally, multi-agent systems can simulate market behaviour with high fidelity, enabling better risk modelling and strategic forecasting (Ardon et al., 2021). These capabilities position game-playing AI not just as a trading assistant, but as a transformative tool for predictive decision-making in complex, fast-moving financial systems.

QN 3. ETHICAL PROBLEMS WITH GAME-PLAYING AI SYSTEMS

I think one of the core ethical issues with game-playing AI systems is not the unfair advantage of the algorithms per se, but how it redefines what it means to be exceptional. As AI sets new performance standards, human achievement can seem less impressive, leading to a credibility bias that undervalues genuine skill—not only when it falls short of AI, but even when it matches it, as people may question whether the achievement is truly human.

Beyond this, there are also ethical concerns about how humans choose to deploy these systems—particularly in financially exploitative ways, such as manipulating in-game economies, targeting vulnerable players, or designing AI-driven monetization loops that prioritise profit over fair play.

3.1 WHEN AI MAKES MASTERS QUESTION THEIR WORTH

Lee Sedol, one of the greatest Go players in history, retired in 2019 specifically because of AI and out of despair. His retirement speaks volume about the profound impact game-playing AI has on human achievement in games-something that extends beyond unfair competition into existential territory. With AI replacing human task performance, this technological win felt like a loss for humankind (Danaher & Nyholm, 2020).

3.2 FINANCIAL EXPLOITATION USING AI GAMING SYSTEMS

Gaming AI systems increasingly target players for financial exploitation. King et al. (2019) examined 13 video game monetization patents and found that almost all relied on exploiting players' data to optimize the delivery and timing of purchase offers. EA was accused of using dynamic difficulty adjustment to manipulate spending, allegedly making games harder to push players toward purchasing advantages (Dring, 2021). While the lawsuit was dismissed, this case highlights how AI can manipulate players' data for profit.

In a similar study, Melhart et al. (2023) found that these systems use affective computing models (Devillers & Cowie, 2023) explicitly designed to exploit psychological vulnerabilities for financial gain by targeting "whales" (high spenders) and emotionally vulnerable players. These models can detect players' emotional states to optimize purchase offer timing, while battle passes create "fear of missing out" to drive spending. These systems also lack transparency and accountability, representing a fundamental threat to player autonomy, particularly for impressionable children.

QN 4. ARE NEURAL NETWORKS THE BEST GAME PLAYERS?

Whether neural networks truly represent the peak of artificial game-playing intelligence is still up for debate. Recent research shows a mixed picture—on one hand, there have been remarkable breakthroughs, but on the other, some key limitations remain.

4.1 THE CASE FOR NEURAL NETWORK

The pattern recognition and learning advantages of neural networks provide fundamental superiority over traditional approaches. Convolutional layers automatically extract hierarchical visual features from game states, while reinforcement learning enables continuous improvement through millions of self-play games. The results of that are novel strategies beyond human knowledge (e.g., AlphaGo's famous winning "Move 37" against Lee Sedol) thanks to the Monte Carlo Tree Search Algorithm (Silver et al., 2016). Neural networks have achieved unprecedented victories in the most complex games known to mankind within the last decade.

4.2 THE CASE AGAINST NEURAL NETWORK

Despite impressive achievements, neural networks still suffer from some limitations in terms of some fundamental learning failures. The Game of Life study based on a "grid-based automaton" shows how neural networks consistently fail to learn simple deterministic rules despite receiving 1 million training examples (Dickson, 2020). This represents a critical weakness: if neural networks cannot master basic logical rules, their apparent success in complex games may reflect overfitting to specific scenarios rather than genuine strategic understanding (Springer & Kenyon, 2020).

4.3 OTHER WELL-PERFORMING MODELS AND HYBRID-MODEL SYSTEM

Recent studies highlight the potential of large language models (LLMs) in gaming, showing their ability to generate reasoning structures and adapt to new scenarios (Hu et al., 2024). LLMs also align closely with human difficulty assessments using simple prompts, avoiding the heavy costs of rule-based systems or traditional ML training (Xiao & Yang, 2014). A major breakthrough is Meta's CICERO, which achieved human-level performance in the strategy game *Diplomacy* by combining LLMs with advanced reasoning, outperforming traditional neural networks in strategic communication and long-term planning (Bakhtin et al., 2022).

THERE IS NO "BEST" MODEL

In conclusion, while neural networks excel in visual tasks, LLMs often outperform them in games that require language understanding, strategic reasoning, and adaptability. Their growing success highlights that performance depends less on model type and more on factors like training methods, data quality, and fine-tuning techniques. Rather than seeking a single "best" model, the focus should be on aligning the right tools and synergy with the right challenges.

QN 5. RELIABILITY OF REFERENCES

5.1 “DOTA 2 WITH LARGE SCALE DEEP REINFORCEMENT LEARNING” – OPENAI ET AL.

This paper, published in 2019 on the widely used preprint platform arXiv, has since garnered over 2,297 citations, including references from leading institutions such as DeepMind and MIT—highlighting its strong influence within the field of game AI. Although it has not undergone formal peer review, its credibility is reinforced by the calibre of its authorship, which includes renowned OpenAI researchers such as Ilya Sutskever, co-founder of OpenAI and a highly respected figure in AI research. The volume of citations and the involvement of prominent contributors compensate for the lack of peer review, making this a reliable and well-regarded source for the purposes of this report.

5.2 “GRANDMASTER LEVEL IN STARCRAFT II USING MULTI-AGENT REINFORCEMENT LEARNING” – VINYALS ET AL.

Published in 2019 in *Nature*—one of the most prestigious and widely respected peer-reviewed scientific journals—this paper has been cited over 5,000 times, including by leading research institutions such as OpenAI, Microsoft Research, and Facebook AI Research. Authored by a 47-member team at DeepMind, the paper features notable contributors like Demis Hassabis, who later co-won the 2024 Nobel Prize in Chemistry for his work on AlphaFold. Given its high citation count, publication venue, and the reputation of its authors, this paper demonstrates exceptional reliability across all major indicators of scientific credibility.

5.3 “HUMAN-LEVEL PLAY IN THE GAME OF DIPLOMACY BY COMBINING LANGUAGE MODELS WITH STRATEGIC REASONING” – BAKHTIN ET AL.

This Meta FAIR paper was published in *Science*, the flagship journal of the Science family of peer-reviewed publications, hosted on the Science.org platform and published by the American Association for the Advancement of Science (AAAS)—one of the most respected scientific organizations globally. Its scientific credibility is reinforced by its authorship, which includes elite AI researchers from Meta’s Fundamental AI Research (FAIR) division, a world-leading industrial AI lab known for collaborations with institutions such as MIT, Carnegie Mellon, and Columbia University. In particular, the team includes Noam Brown, recognized for developing the AI systems *Libratus* and *Pluribus*, which defeated professional world champions in poker. Together, the journal’s prestige, institutional backing, and the researchers’ proven track record strongly affirm the paper’s scientific reliability.

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